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Q2 Sketch these lines & decide if the eqⁿs are solvable:

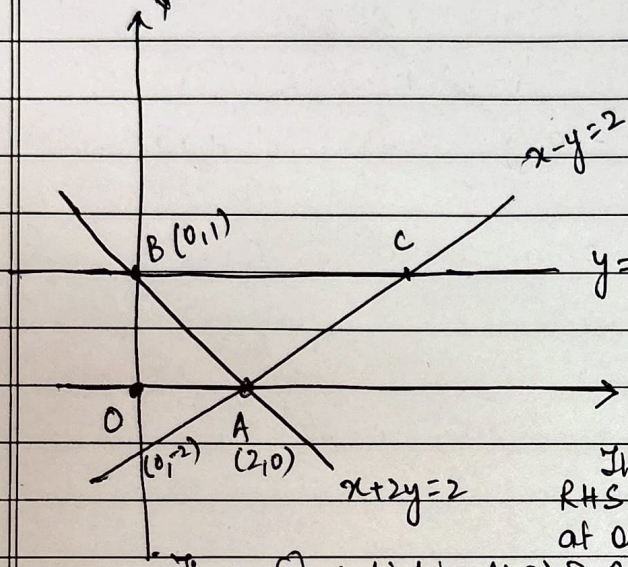
$x + 2y = 2$ — ①

$x - y = 2$ — ②

$y = 1$ — ③

What happens if all RHS are zero? Is there any non-zero choice of RHS that allows the three lines to intersect at the same point.

Sol^{n}



The sys. of eqⁿs is not solvable as a whole ∵ they are overdetermined (more eqⁿs than no. of unknowns)

If all RHS are zero

Then $x=0, y=0$ is the solution.

There can be infinite such RHS so that they can intersect at one pt. Say pt. of int. be (1, 2)

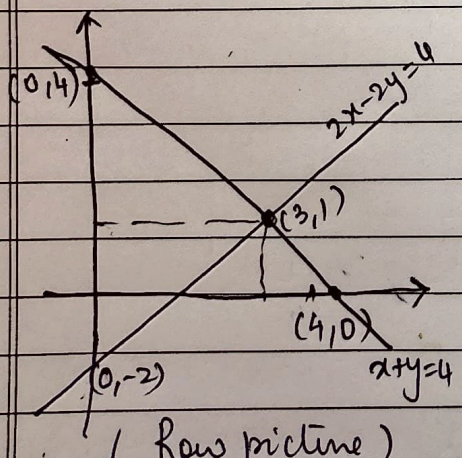
Then, ① satisfies (1,2) & RHS will be (5) for ② RHS will be (−1) & for ③ RHS will be (2)

Q3:- For the eqⁿs $x + y = 4$

$2x - 2y = 4$ draw the row picture (two intersecting lines)

& a column picture (comb. of 2 columns = to the col. vector (4,4) on the rt. side).

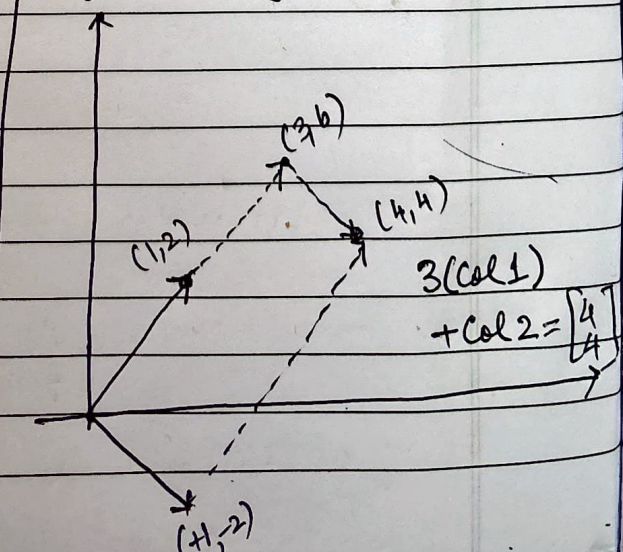
Sol^{n}



(Row picture)

(Col Picture)

$$x\begin{bmatrix}1\\2\end{bmatrix} + y\begin{bmatrix}1\\-2\end{bmatrix} = \begin{bmatrix}4\\4\end{bmatrix}$$



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Problems to be solved from Prob. Set 1.2

Q2, 3, 5, 6, 7, 11

Q5:- Describe the intersection of the 3 planes
$u+v+w+z=6$ & $u+w+z=4$
$u+w=2$ (all in 4-D space).

Is it a line or a point or an empty set?
What is the intersection if the 4th plane $u=-1$ is included?

Soln

$$u+v+w+z=6 \quad \text{—①}$$
$$u+w+z=4 \quad \text{—②}$$
$$u+w=2 \quad \text{—③}$$

Subs. ③ in ② $\quad 2+z=4 \Rightarrow z=2.$

Subs. ③ in ① & taking $z=2$ in ①

$$2+v+2=6 \Rightarrow v=2$$

Since 4 variables & 3 eqn, of which
$v=2, z=2$
So consider $u=u$.
$\therefore w=2-u.$
$v=2$
$z=2.$

This means that the intersection of the 3 planes can be expressed as

(A line requires one free parameter, in this case u that can vary & thus a line)

$(u, v, w, z) = (u, 2, 2-u, 2)$

where u can take any real value.

This describes a line in $\mathbb{R}^4$.

(The intersection of 3 planes describe a line in $\mathbb{R}^4$)
(A plane requires at least 2 free parameters).

Including 4th plane, $u = -1$.

We get $v = 2$, $w = 2 - (-1) = 2 + 3$, $z = 2$.

Thus, the intersection of 4 planes is the single point $(-1, 2, 3, 2)$

(no free parameter)

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Qb:- These eq^ns are certain to have a solution $x=y=0$. For which values of a is there a whole line of solutions?

$$ax + 2y = 0$$
$$2x + ay = 0$$

Sol^n For having infinite no. of solutions the eq^ns must be dependent,

ie one can be expressed as a multiple of the other.

For finding the values of a for which these eq^ns are dependent, set the coeff.s proportional:

$$\frac{a}{2} = \frac{2}{a} \Rightarrow a^2 = 4$$
$$\text{or } a = \pm 2$$

Or This can be obtained as.

$$2(ax + 2y = 0) \rightarrow 2ax + 4y = 0$$
$$a(2x + ay = 0) \quad 2ax + a^2y = 0$$

$$(4 - a^2)y = 0$$

As $y=0$ is already a stated sol, so, we take

$$4 - a^2 = 0 \Rightarrow a = \pm 2$$

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Q7. Explain why

$$u+v+w=2 \quad \text{---} (1)$$
$$u+2v+3w=1 \quad \text{---} (2)$$
$$v+2w=0 \quad \text{---} (3).$$

is singular by finding a combination of the three eqns that adds up to $0=1$. What value should replace last zero on RHS to allow the eqns to have solution & what is one of the sols?

Sol^n $(1)+(3) \Rightarrow u+2v+3w=2$
& (2) is $u+2v+3w=1$.

$\therefore$ $(1)+(3)-(2)$ gives $0=1$

In the last eq^n (3), zero should be replaced by -1 so that it becomes $0=0$ (infinitely many solutions)

$(4,-3,1)$ is one such solution.

Q11:- The column picture for Q7 (singular sys) is

$$u\begin{bmatrix}1\\1\\0\end{bmatrix}+v\begin{bmatrix}1\\2\\1\end{bmatrix}+w\begin{bmatrix}1\\3\\2\end{bmatrix}=b$$

Show that the three columns on the left lie in the same plane by expressing the third column as a combination of the first two. What are all solutions (u,v,w) if b is a zero vector $[0,0,0)$?

88.

Taking ~~$u=4, v=-3, w=1$~~

$$u\begin{bmatrix}1\\1\\0\end{bmatrix} + v\begin{bmatrix}1\\2\\1\end{bmatrix} + w\begin{bmatrix}1\\3\\2\end{bmatrix} = b.$$

Firstly, we'll show that.

$$\begin{bmatrix}1\\3\\2\end{bmatrix} \text{ is a lc of } \begin{bmatrix}1\\1\\0\end{bmatrix} + \begin{bmatrix}1\\2\\1\end{bmatrix}$$

$\Rightarrow$ the 3 ~~planes~~ columns lie in the same plane.

$$(-1)\begin{bmatrix}1\\1\\0\end{bmatrix} + 2\begin{bmatrix}1\\2\\1\end{bmatrix} = \begin{bmatrix}1\\3\\2\end{bmatrix}$$

$\therefore$ The 3 columns lie in the same plane.

Now, given that $b = \begin{bmatrix}0\\0\\0\end{bmatrix}$.

$$\Rightarrow u\begin{bmatrix}1\\1\\0\end{bmatrix} + v\begin{bmatrix}1\\2\\1\end{bmatrix} + w\begin{bmatrix}1\\3\\2\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

or
$$u+v+w=0 \quad \text{——} (1)$$
$$u+2v+3w=0 \quad \text{——} (2)$$
$$v+2w=0 \quad \text{——} (3)$$

from (3), $v=-2w$

Subs. in (2) $u-4w+3w=0$ or $u-w=0$

$\Rightarrow u=w$

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$$\therefore (u, v, w) = (w, -2w, w)$$

$$= w(1, -2, 1)$$

$$= \text{span}\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}$$

w can take any value & we'll get Corresp. values of u & v.

Solutions for MFAD Unit 1 Classwork Problems (Gaussian Elimination)

1. For which numbers a does elimination break down (a) permanently and (b) temporarily?

$$ax + 3y = -3$$
$$4x + 6y = 6.$$

Solve for x and y after fixing the second breakdown by row exchange.

Solution:

(a) When the above system of equations is expressed in augmented form, the resultant matrix is

$$[A|b] = \left[\begin{array}{cc|c} a & 3 & -3 \\ 4 & 6 & 6 \end{array}\right]$$

In order to convert the augmented matrix $[A \mid b]$ into echelon form, we have to reduce the matrix in such a way that the element below the first pivot element (in this case, a) is zero.

$$R_2 \to R_2 - \frac{4}{a}R_1$$

Thus, the resultant matrix is

$$[U|c] = \left[\begin{array}{cc|c} a & 3 & -3 \\ 0 & 6 - \frac{12}{a} & 6 + \frac{12}{a} \end{array}\right]$$

In order for elimination to break down, either one of the pivots (i.e., elements along the principal diagonal of the matrix U) must be equal to zero. This happens when $a = 0$ or $a = 2$. One of the values thus obtained results in permanent breakdown, and the other results in breakdown which is temporary and restorable.

Substituting $a = 2$ in the original matrix, we get

$$\left[\begin{array}{cc|c} 2 & 3 & -3 \\ 4 & 6 & 6 \end{array}\right]$$

Converting to echelon form after executing the row operation $R_2 \to R_2 - 2R_1$ yields

$$\left[\begin{array}{cc|c} 2 & 3 & -3 \\ 0 & 0 & 12 \end{array}\right]$$

The above system of equations is inconsistent, i.e., has no solution. This inconsistency cannot be fixed using a row exchange. Since the second equation produces $0 = 12$ regardless of row order, the system remains contradictory.

Therefore, $a = 2$ leads to a permanent breakdown of elimination.

(b) As explored in the solution for subproblem (a), we arrive at two numbers for which elimination breakdown happens, i.e., $a = 0$ and $a = 2$.

Taking $a = 0$ in the original matrix, we get

$$\left[\begin{array}{cc|c} 0 & 3 & -3 \\ 4 & 6 & 6 \end{array}\right]$$

The first pivot element in the matrix is zero, which may seemingly make it look like we're stuck in this problem as we cannot find an appropriate multiplier to eliminate x in R_2 using R_1. The good news is, we can fix this problem by using row exchanges.
We swap the positions of the rows R_1 and R_2 in order to convert the matrix into echelon form. Thus, our resultant matrix after the row exchange operation

$$R_1 \leftrightarrow R_2,$$

is

$$\left[\begin{array}{cc|c} 4 & 6 & 6 \\ 0 & 3 & -3 \end{array}\right]$$

We can see that this matrix is already in echelon form, and no further elimination is required. We obtain the following equations after elimination:

$$\begin{aligned} 4x + 6y &= 6 \\ 0x + 3y &= -3. \end{aligned}$$

Therefore, the solutions x and y thus obtained are $x = 3$ and $y = -1$ respectively.

■

2. Which number q makes this system singular and which right-hand side t gives it infinitely many solutions? Find the solution that has $z = 1$.

$$\begin{aligned} x + 4y - 2z &= 1 \\ x + 7y - 6z &= 6 \\ 3y + qz &= t. \end{aligned}$$

Solution:

(a) Writing the above system of equations in augmented form, we get

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 1 & 7 & -6 & 6 \\ 0 & 3 & q & t \end{array}\right]$$

To eliminate x from the second row, we subtract R_1 from R_2.

$$R_2 \to R_2 - R_1$$

Our resultant matrix thus becomes

$$\left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 0 & 3 & -4 & 5 \\ 0 & 3 & q & t \end{array}\right]$$

To eliminate y from the third row, we subtract R_2 from R_3.

$$R_3 \to R_3 - R_2$$

Thus, our resultant matrix is

$$[U|c] = \left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 0 & 3 & -4 & 5 \\ 0 & 0 & q+4 & t-5 \end{array}\right]$$

In order for the system of equations to be singular, one of the pivots in U must be zero. Two of the pivots, i.e., the pivots associated with x and y are non-zero, therefore, only one pivot, i.e., the pivot associated with z can be zero if $q + 4 = 0$. Therefore, $q = -4$ makes the system of equations singular.

In order for the system of equations to have infinitely many solutions, the ranks of the matrix U and the matrix $[U|c]$ must be equal and lower than the number of free variables in the system (three in this case). If $q = -4$, the rank of the matrix U is thus 2 and so the rank of the matrix $[U|c]$

must be 2 as well. Therefore, in order for this to hold true, $t - 5 = 0$ and thus $t = 5$.

Therefore, for infinitely many solutions, $q = -4$ and $t = 5$.

(b) Substituting $z = 1$, we get

$$\begin{aligned} x + 4y - 2 &= 1 \\ x + 7y - 6 &= 6 \\ 3y + q &= t. \end{aligned}$$

The system of equations in augmented form thus becomes

$$[A|b] = \left[\begin{array}{cc|c} 1 & 4 & 3 \\ 1 & 7 & 12 \\ 0 & 3 & 9 \end{array}\right]$$

since $t = 5$, $q = -4$ and thus $t - q = 9$.

Eliminating x from R_2 by subtracting R_1 from R_2, we get

$$R_2 \to R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & 3 & 9 \\ 0 & 3 & 9 \end{array}\right]$$

Eliminating y from R_3 by subtracting R_2 from R_3, we get

$$R_3 \to R_3 - R_2$$

$$[U|c] = \left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & 3 & 9 \\ 0 & 0 & 0 \end{array}\right]$$

Solving the resultant system of equations, we get $y = 3$ and $x = -9$.

Thus, the solution is $(x, y, z) = (-9, 3, 1)$

■

3. Apply elimination and back-substitution to solve

$$\begin{aligned} 2u + 3v &= 0 \\ 4u + 5v + w &= 3 \\ 2u - v - 3w &= 5. \end{aligned}$$

What are the pivots? List the three operations in which a multiple of one row is subtracted from another.

Solution:

Expressing the above system of equations in augmented form, we get

$$[A|b] = \left[\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ 4 & 5 & 1 & 3 \\ 2 & -1 & -3 & 5 \end{array}\right]$$

Eliminating u from the second row and third row, we get

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ 0 & -1 & 1 & 3 \\ 0 & -4 & -3 & 5 \end{array}\right]$$

Eliminating v from the third row, we get

$$R_3 \to R_3 - 4R_2$$

$$[U|c] = \left[\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & -7 & -7 \end{array}\right]$$

Looking at the above matrix, we can see that the pivots thus obtained are 2, -1 and 1.

Using back-substitution, we get $-7w = -7 \implies w = 1$.

Using $w = 1$ in the equation $-v + w = 3$, we get $v = -2$.

Using $v = -2$ in the equation $2u + 3v = 0$, we get $u = 3$.

Therefore, the solution thus obtained is $(u, v, w) = (3, -2, 1)$.

■

4. Solve the system to find pivots when

$$\begin{aligned} 2u - v &= 0 \\ -u + 2v - w &= 0 \\ -v + 2w - z &= 0 \\ -w + 2z &= 5. \end{aligned}$$

You may carry the right hand side as a fifth column (and omit writing u, v, w, z until the solution at the end).

Solution:

The above system of equations can be written in the form

$$[A|b] = \left[\begin{array}{cccc|c} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & 5 \end{array}\right]$$

Eliminating u from the second row, we get

$$R_2 \to R_2 + \frac{1}{2}R_1$$

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & 5 \end{array}\right]$$

Eliminating v from the third row, we get

$$R_3 \to R_3 + \frac{2}{3}R_2$$

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -1 & 0 \\ 0 & 0 & -1 & 2 & 5 \end{array}\right]$$

Eliminating w from the fourth row, we get

$$R_4 \to R_4 + \frac{3}{4}R_3$$

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -1 & 0 \\ 0 & 0 & 0 & \frac{5}{4} & 5 \end{array}\right]$$

The pivots thus obtained are 2, $\frac{3}{2}$, $\frac{4}{3}$ and $\frac{5}{4}$.

Using back-substitution to solve these equations, we get $z = 4$, $w = 3$, $v = 2$ and $u = 1$.

Therefore, the solution is $(u, v, w, z) = (1, 2, 3, 4)$.

■

5. Use elimination to solve

$$\begin{aligned} u + v + w &= 6 \\ u + 2v + 2w &= 11 \\ 2u + 3v - 4w &= 3 \end{aligned}$$

and

$$\begin{aligned} u + v + w &= 7 \\ u + 2v + 2w &= 10 \\ 2u + 3v - 4w &= 3 \end{aligned}$$

Solution:

We express the first system of equations in augmented form.

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 2 & 11 \\ 2 & 3 & -4 & 3 \end{array}\right]$$

Eliminating u from the second row and from the third row, we get

$$R_2 \to R_2 - R_1$$

$$R_3 \to R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 1 & -6 & -9 \end{array}\right]$$

Eliminating v from the third row, we get

$$R_3 \to R_3 - R_2$$

$$[U|c] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & -7 & -14 \end{array}\right]$$

Using back-substitution to solve these equations, we get $w = 2$, $v = 3$ and $u = 1$.

The solution thus obtained is $(u, v, w) = (1, 3, 2)$.

We express the second system of equations in augmented form.

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 1 & 2 & 2 & 10 \\ 2 & 3 & -4 & 3 \end{array}\right]$$

Eliminating u from the second row and from the third row, we get

$$R_2 \to R_2 - R_1$$

$$R_3 \to R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -6 & -11 \end{array}\right]$$

Eliminating v from the third row, we get

$$R_3 \to R_3 - R_2$$

$$[U|c] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -7 & -14 \end{array}\right]$$

Using back-substitution to solve these equations, we get $w = 2$, $v = 1$ and $u = 4$.

The solution thus obtained is $(u, v, w) = (4, 1, 2)$.

■

6. For which three numbers a will elimination fail to give three pivots?

$$\begin{aligned} ax + 2y + 3z &= b_1 \\ ax + ay + 4z &= b_2 \\ ax + ay + az &= b_3 \end{aligned}$$

Solution:

We express the above system of equations in augmented form.

$$[A|b] = \left[\begin{array}{ccc|c} a & 2 & 3 & b_1 \\ a & a & 4 & b_2 \\ a & a & a & b_3 \end{array}\right]$$

Eliminating x from the second and third row, we get

$$R_2 \to R_2 - R_1$$

$$R_3 \to R_3 - R_1$$

$$\left[\begin{array}{ccc|c} a & 2 & 3 & b_1 \\ 0 & a-2 & 1 & b_2 - b_1 \\ 0 & a-2 & a-3 & b_3 - b_1 \end{array}\right]$$

Eliminating y in the third row, we get

$$R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|c} a & 2 & 3 & b_1 \\ 0 & a-2 & 1 & b_2 - b_1 \\ 0 & 0 & a-4 & b_3 - b_2 \end{array}\right]$$

Elimination breakdown occurs if any one of the pivots a, $a-2$ or $a-4$ becomes zero.

Therefore, the three values of a for which elimination will break down are 0, 2 and 4.

■

Solutions for MFAD Unit 1 Classwork Problems (Elimination Matrices)

1. Which three matrices E_{21}, E_{31} and E_{32} put A into triangular form U?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix}$$

and $E_{32}E_{31}E_{21}A = U$.

Multiply those E's to get one matrix M that does elimination: $MA = U$

Solution:

In order to figure out the elimination matrices, we have to convert A into echelon form U. At each row operation, we record the multiplier and use it in our answer.

Eliminating x from R_2 and R_3, we get

$$R_2 \to R_2 - 4R_1$$

$$R_3 \to R_3 + 2R_1$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 4 & 0 \end{bmatrix}$$

The multiplier used in order to eliminate x from R_2 using R_1 was $l_{21} = 4$. The multiplier used in order to eliminate x from R_3 using R_1 was $l_{31} = -2$.

Eliminating y from R_3, we get

$$R_3 \to R_3 - 2R_2$$

$$U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

The multiplier used in order to eliminate y from R_3 using R_2 was $l_{32} = 2$.

The first elimination matrix E_{21} thus becomes

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The second elimination matrix E_{31} thus becomes

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -l_{31} & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

The third elimination matrix E_{32} thus becomes

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -l_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

Multiplying these matrices, we get

$$E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix} = M \tag{1}$$

Multiplying the matrix M with A, we get

$$MA = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix} = U \tag{2}$$

Thus, $E_{32}E_{31}E_{21}A = MA = U$ has been verified.

■

2. This 4 by 4 matrix needs which elimination matrices E_{21}, E_{32} and E_{43}?

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Solution:

We have to convert the matrix A into its echelon form U. We assume that the free variables are w, x, y and z.

Eliminating w from R_2, we get

$$R_2 \rightarrow R_2 + \frac{1}{2}R_1$$

$$\begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

The multiplier used in order to eliminate w from R_2 using R_1 was $l_{21} = -\frac{1}{2}$.

Eliminating x from R_3, we get

$$R_3 \rightarrow R_3 + \frac{2}{3}R_2$$

$$\begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & 0 & \frac{4}{3} & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

The multiplier used in order to eliminate x from R_3 using R_2 was $l_{32} = -\frac{2}{3}$.

Eliminating y from R_4, we get

$$R_4 \rightarrow R_4 + \frac{3}{4}R_3$$

$$U = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 \\ 0 & 0 & \frac{4}{3} & -1 \\ 0 & 0 & 0 & \frac{5}{4} \end{bmatrix}$$

The multiplier used in order to eliminate y from R_4 using R_3 was $l_{43} = -\frac{3}{4}$.

The first elimination matrix E_{21} thus becomes

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -l_{21} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The first elimination matrix E_{32} thus becomes

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -l_{32} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{2}{3} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The third elimination matrix E_{43} thus becomes

$$E_{43} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -l_{43} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{3}{4} & 1 \end{bmatrix}$$

■

Triangular Factors and Row Exchanges

Problem set 1.5

Problem 1

Apply elimination to produce the factors L and U for

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 4 \\ 1 & 4 & 8 \end{bmatrix}.$$

Solution for i)

Matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

To eliminate the entry in row 2, column 1, subtract 4 times row 1 from row 2 ($R_2 \to R_2 - 4R_1$). The multiplier l_{21} is 4.

This gives us our upper triangular matrix U:

$$U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

Place the multiplier into the corresponding position in the identity matrix to form L:

$$L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

Solution for ii)

Matrix:

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

To eliminate the entries in column 1 below the pivot (3), subtract $\frac{1}{3}$ times row 1 from row 2 ($R_2 \to R_2 - \frac{1}{3}R_1$), and subtract $\frac{1}{3}$ times row 1 from row 3 ($R_3 \to R_3 - \frac{1}{3}R_1$). The multiplier l_{21} is $\frac{1}{3}$ and the multiplier l_{31} is $\frac{1}{3}$. The intermediate matrix is:

$$A' = \begin{bmatrix} 3 & 1 & 1 \\ 0 & \frac{8}{3} & \frac{2}{3} \\ 0 & \frac{2}{3} & \frac{8}{3} \end{bmatrix}$$

To eliminate the entry in column 2 below the new pivot ($\frac{8}{3}$), subtract $\frac{1}{4}$ times row 2 from row 3 ($R_3 \to R_3 - \frac{1}{4}R_2$). The multiplier l_{32} is $\frac{1}{4}$.

This yields our upper triangular matrix U:

$$U = \begin{bmatrix} 3 & 1 & 1 \\ 0 & \frac{8}{3} & \frac{2}{3} \\ 0 & 0 & \frac{5}{2} \end{bmatrix}$$

Place the multipliers into the corresponding positions to form L:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{bmatrix}$$

Solution for iii)

Matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 4 \\ 1 & 4 & 8 \end{bmatrix}$$

To eliminate the entries in column 1 below the pivot (1), subtract row 1 from row 2 ($R_2 \to R_2 - 1R_1$), and subtract row 1 from row 3 ($R_3 \to R_3 - 1R_1$). The multiplier l_{21} is 1 and the multiplier l_{31} is 1. The intermediate matrix is:

$$A' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 3 \\ 0 & 3 & 7 \end{bmatrix}$$

To eliminate the entry in column 2 below the new pivot (3), subtract row 2 from row 3 ($R_3 \to R_3 - 1R_2$). The multiplier l_{32} is 1.

This yields our upper triangular matrix U:

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

Place the multipliers into the corresponding positions to form L:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Problem 2

Which numbers a, b, c lead to row exchanges? Which make the matrix singular?

$$A = \begin{bmatrix} 1 & 2 & 0 \\ a & 8 & 3 \\ 0 & b & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} c & 2 \\ 6 & 4 \end{bmatrix}.$$

Solution for i)

$$A = \begin{bmatrix} 1 & 2 & 0 \\ a & 8 & 3 \\ 0 & b & 5 \end{bmatrix}$$

To eliminate the entry in row 2, column 1, subtract a times row 1 from row 2 ($R_2 \to R_2 - aR_1$). The multiplier l_{21} is a. The intermediate matrix is:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 8-2a & 3 \\ 0 & b & 5 \end{bmatrix}$$

A row exchange is required if the second pivot is zero, which happens when $8 - 2a = 0$, meaning $a = 4$. Assuming $a \neq 4$, to eliminate the entry in row 3, column 2, subtract $\frac{b}{8-2a}$ times row 2 from row 3 ($R_3 \to R_3 - \frac{b}{8-2a}R_2$). The multiplier l_{32} is $\frac{b}{8-2a}$. This yields the upper triangular matrix:

$$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 8-2a & 3 \\ 0 & 0 & 5 - \frac{3b}{8-2a} \end{bmatrix}$$

The matrix is singular if the third pivot is zero, which means $5 - \frac{3b}{8-2a} = 0$. Solving this gives $40 - 10a - 3b = 0$, or $3b + 10a = 40$. The matrix is also singular if $a = 4$ and $b = 0$.

Solution for ii)

$$A = \begin{bmatrix} c & 2 \\ 6 & 4 \end{bmatrix}$$

A row exchange is required if the first pivot is zero, which means $c = 0$. Assuming $c \neq 0$, to eliminate the entry in row 2, column 1, subtract $\frac{6}{c}$ times row 1 from row 2 ($R_2 \to R_2 - \frac{6}{c}R_1$). The multiplier l_{21} is $\frac{6}{c}$. This yields the upper triangular matrix:

$$U = \begin{bmatrix} c & 2 \\ 0 & 4 - \frac{12}{c} \end{bmatrix}$$

The matrix is singular if the second pivot is zero, which means $4 - \frac{12}{c} = 0$, simplifying to $c = 3$.

Problem 3

Find the $PA = LDU$ factorizations (and check them) for

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution for i)

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

A row exchange is needed immediately because the first pivot is zero. Swap row 1 and row 2. The permutation matrix P is the identity matrix with its first two rows swapped. The permuted matrix PA is:

$$PA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

To eliminate the entry in row 3, column 1, subtract 2 times row 1 from row 3 ($R_3 \to R_3 - 2R_1$). The multiplier l_{31} is 2. The intermediate matrix is:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

To eliminate the entry in row 3, column 2, subtract 3 times row 2 from row 3 ($R_3 \to R_3 - 3R_2$). The multiplier l_{32} is 3. The resulting upper triangular matrix is:

$$U' = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

Factoring out the diagonal pivots into matrix D yields the $PA = LDU$ factorization:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution for ii)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

To eliminate the entries in column 1, subtract 2 times row 1 from row 2 ($R_2 \to R_2 - 2R_1$), and subtract 1 times row 1 from row 3 ($R_3 \to R_3 - 1R_1$). The multipliers are $l_{21} = 2$ and $l_{31} = 1$. The intermediate matrix is:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

The pivot in row 2 is missing (it is zero). A row exchange is required, so we swap row 2 and row 3. The permutation matrix P swaps rows 2 and 3 of the identity matrix. Applying this to the multipliers as well, l_{21} becomes 1 and l_{31} becomes 2. The permuted upper triangular matrix is:

$$U' = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Factoring out the diagonal pivots into matrix D yields the $PA = LDU$ factorization:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Problem 4

Compute L and U for the symmetric matrix

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}.$$

Find four conditions on a, b, c, d to get $A = LU$ with four pivots.

Solution

For the symmetric matrix,

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$$

To eliminate the entries in column 1, subtract 1 times row 1 from row 2, row 3, and row 4. The multipliers l_{21}, l_{31}, and l_{41} are all 1. The intermediate matrix is:

$$\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix}$$

To eliminate the entries in column 2, subtract 1 times row 2 from row 3 and row 4. The multipliers l_{32} and l_{42} are both 1. The intermediate matrix is:

$$\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$$

To eliminate the entry in column 3, subtract 1 times row 3 from row 4. The multiplier l_{43} is 1. This yields the upper triangular matrix U:

$$U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

Consolidating all the multipliers, we get the lower triangular matrix L:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

For the matrix A to have an LU decomposition with four pivots, all four diagonal entries of U must be non-zero. The four conditions are $a \neq 0$, $b - a \neq 0$, $c - b \neq 0$, and $d - c \neq 0$. Therefore, the four conditions are $a \neq 0$, $b \neq a$, $c \neq b$, and $d \neq c$.

Inverses and Transposes

Problem Set 1.6

Problem 1

Question: Use the Gauss-Jordan method to invert

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution for i)

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

To find the inverse using Gauss-Jordan elimination, we form the augmented matrix $[A_1 \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

To eliminate the entry in row 2, column 1, subtract 1 times row 1 from row 2 ($R_2 \to R_2 - 1R_1$). The multiplier l_{21} is 1. The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

Now, eliminate upwards. To eliminate the entry in row 2, column 3, subtract 1 times row 3 from row 2 ($R_2 \to R_2 - 1R_3$). The resulting augmented matrix has the identity matrix on the left:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

The inverse matrix A_1^{-1} is on the right:

$$A_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution for ii)

$$A_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

We form the augmented matrix $[A_2 \mid I]$:

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

To eliminate the entry in row 2, column 1, subtract $-\frac{1}{2}$ times row 1 from row 2 ($R_2 \to R_2 - (-\frac{1}{2})R_1$). The multiplier l_{21} is $-\frac{1}{2}$. The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

To eliminate the entry in row 3, column 2, subtract $-\frac{2}{3}$ times row 2 from row 3 ($R_3 \to R_3 - (-\frac{2}{3})R_2$). The multiplier l_{32} is $-\frac{2}{3}$. The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array}\right]$$

Now, eliminate upwards. To eliminate the entry in row 2, column 3, subtract $-\frac{3}{4}$ times row 3 from row 2 ($R_2 \to R_2 - (-\frac{3}{4})R_3$). The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array}\right]$$

To eliminate the entry in row 1, column 2, subtract $-\frac{2}{3}$ times row 2 from row 1 ($R_1 \to R_1 - (-\frac{2}{3})R_2$). The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array}\right]$$

Finally, divide each row by its pivot (row 1 by 2, row 2 by $\frac{3}{2}$, row 3 by $\frac{4}{3}$) to obtain the identity matrix on the left. The inverse matrix A_2^{-1} is on the right:

$$A_2^{-1} = \begin{bmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{bmatrix}$$

Solution for iii)

$$A_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

We form the augmented matrix $[A_3 \mid I]$:

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

A row exchange is needed immediately because the first pivot is zero. Swap row 1 and row 3. The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

The matrix on the left is already in upper triangular form. Now, eliminate upwards. To eliminate the entry in row 2, column 3, subtract 1 times row 3 from row 2 ($R_2 \to R_2 - 1R_3$). The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

To eliminate the entry in row 1, column 3, subtract 1 times row 3 from row 1 ($R_1 \to R_1 - 1R_3$). The intermediate matrix is:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

To eliminate the entry in row 1, column 2, subtract 1 times row 2 from row 1 ($R_1 \to R_1 - 1R_2$). This yields the identity matrix on the left. The inverse matrix A_3^{-1} is on the right:

$$A_3^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Problem 2

Question: Under what conditions on their entries are A and B invertible?

$$A = \begin{bmatrix} a & b & c \\ d & e & 0 \\ f & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & e \end{bmatrix}$$

Solution

For matrix A, a matrix is invertible if it has a full set of non-zero pivots. By performing a row exchange, swapping row 1 and row 3, the matrix becomes a lower triangular matrix:

$$\begin{bmatrix} f & 0 & 0 \\ d & e & 0 \\ a & b & c \end{bmatrix}$$

The pivots are on the main diagonal: f, e, and c. For the matrix to be invertible, all pivots must be non-zero. Therefore, the conditions for A to be invertible are $c \neq 0$, $e \neq 0$, and $f \neq 0$.

For matrix B, the matrix is block diagonal. We can determine its invertibility by examining its pivots during elimination. The third pivot is simply e, which must be non-zero. For the top-left 2×2 block:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

assuming $a \neq 0$, to eliminate the entry in row 2, column 1, subtract $\frac{c}{a}$ times row 1 from row 2 ($R_2 \to R_2 - \frac{c}{a}R_1$). The multiplier l_{21} is $\frac{c}{a}$. The second pivot becomes $d - \frac{bc}{a} = \frac{ad-bc}{a}$. For this pivot to be non-zero, we must have $ad - bc \neq 0$.
If $a = 0$, a row exchange is required, meaning $c \neq 0$ and $b \neq 0$ for pivots to exist, which also implies $ad - bc \neq 0$.
Therefore, the conditions for B to be invertible are $ad - bc \neq 0$ and $e \neq 0$.